Probability and Statistics

Lecture 27-28

Hypothesis Testing

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Hypothesis Testing

- A statistical method to test a hypothesis
  - A hypothesis a statement about a population parameter
  - The test is concerning the estimated value of the parameter
- Objective of the test is to reject
  - The null hypothesis
  - if data is not consistent with the statement

Hypothesis Testing

- Null Hypothesis: Statement that the test tries to reject
- Alternative Hypothesis: Completement of the null hypothesis and is accepted when the null hypothesis is rejected

Two-sided hypothesis testing:

Null Hypothesis: $H_0: \mu = \mu_0$

Alternative Hypothesis: $H_1: \mu \neq \mu_0$

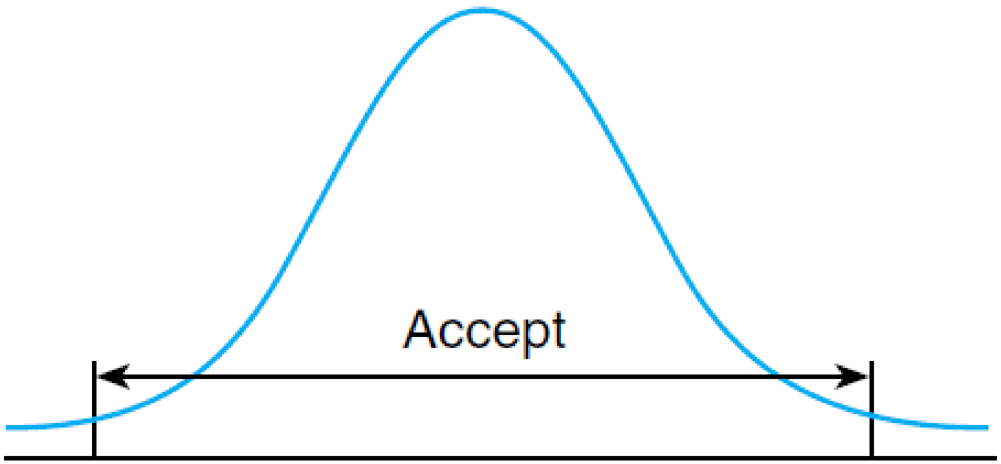
One-sided hypothesis testing:

$$H_0: \mu \leq \mu_0; H_1: \mu > \mu_0$$

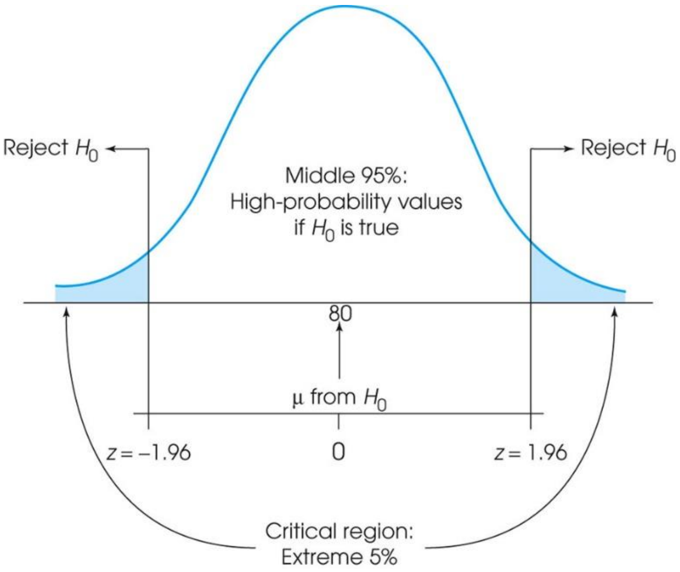
$$H_0: \mu \geq \mu_0; H_1: \mu < \mu_0$$

Decision	H_0 Is True	H_0 Is False
Fail to reject H_0	No error	Type II error
Reject H_0	Type I error	No error

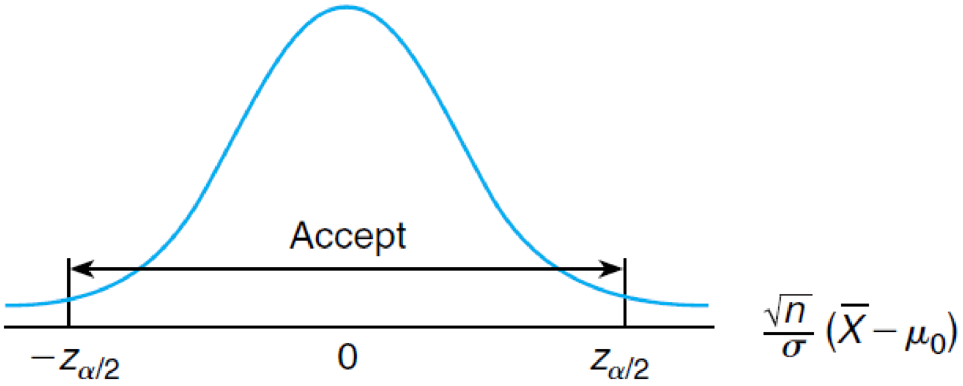
How to test?



Accept



Reject H_0
Reject H_0
Middle 95%:
High-probability values
if H_0 is true
80
μ from H_0
$z = -1.96$
0
$z = 1.96$
Critical region:
Extreme 5%



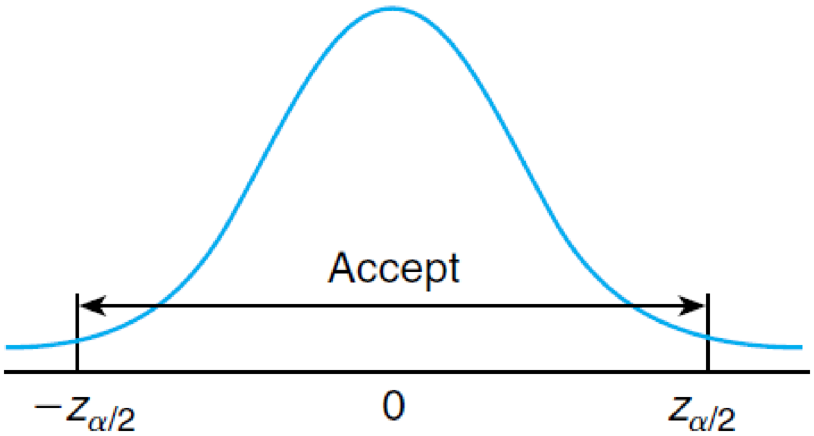
Accept
$-z_{\alpha/2}$
0
$z_{\alpha/2}$
$\frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0)$

Steps to Test Hypothesis

- Identify the parameter of interest
- Specify the test statement and define
  - The null hypothesis H_0
  - The alternative hypothesis H_1
- Specify the MLE of the parameter and determine its distribution
- Determine the test statistic
- State the rejection criteria (either α is given or assume)
  - Determine the critical region
  - Find the critical values
- Collect the sample and calculate
  - MLE of the parameter and test statistic
  - Critical values
- Draw conclusion – whether to reject or not reject H_0

Test for the Mean of Normal Population

Variance is known: Two-sided (two-tailed)



$H_0: \mu = \mu_0$; $H_1: \mu \neq \mu_0$

$\bar{X}$ is the MLE of parameter μ

Test statistic, $T_s = \frac{\overline{X - \mu_0}}{\sigma/\sqrt{n}}$

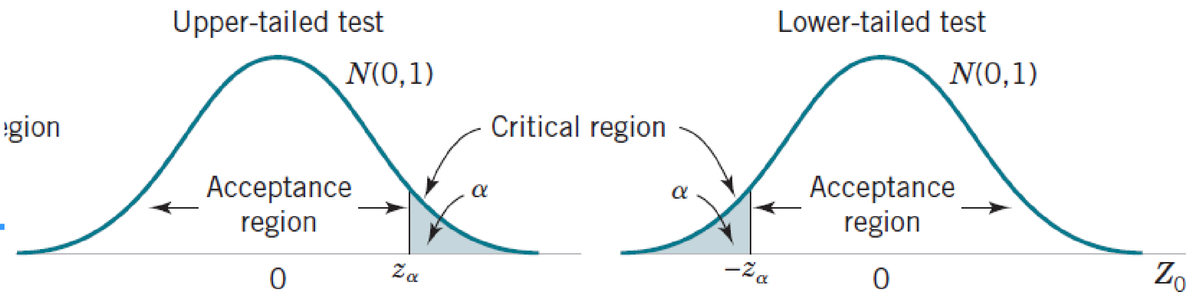
Critical Value: $C_v = z_{\alpha/2}$

Rejection Criteria: if $|T_s| > z_{\alpha/2}$, Reject H_0

Test for the Mean of Normal Population

Variance is known: one-sided (two-tailed)

$H_0: \mu \leq \mu_0; \quad H_1: \mu > \mu_0$ $\bar{X}$ **is the MLE of parameter** μ **Test statistic,** $T_s = \frac{\overline{X - \mu_0}}{\sigma/\sqrt{n}}$ **Critical Value:** $C_v = z_\alpha$ **Rejection Criteria: if** $T_s > z_\alpha$ **, Reject** H_0	$H_0: \mu \geq \mu_0; \quad H_1: \mu < \mu_0$ $\bar{X}$ **is the MLE of parameter** μ **Test statistic,** $T_s = \frac{\overline{X - \mu_0}}{\sigma/\sqrt{n}}$ **Critical Value:** $C_v = z_\alpha$ **Rejection Criteria: if** $T_s < z_\alpha$ **, Reject** H_0



p-value of the test

- How do we define the rejection criteria?
  - Significance level α
  - Critical value $z_{\alpha/2}$ or z_α
- How close is T_s to the critical value?
- What happens if α changes?

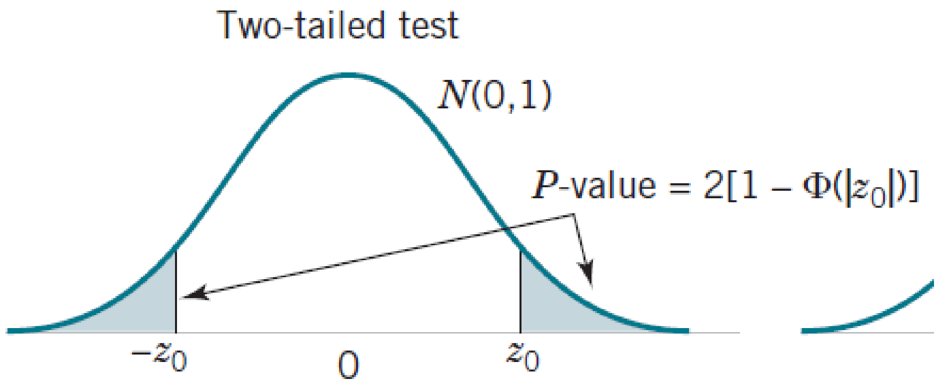
- Significance level is an AUC and we convert it to the critical value to compare it with the test score

Alternative:

- What if we convert the test score its corresponding AUC

p-value of the test

- The p-value is the smallest level of significance that would lead to the rejection of the null hypothesis



EXAMPLE 8.3b In Example 8.3a, suppose that the average of the 5 values received is $\overline{X} = 8.5$. In this case,

$$\frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| = \frac{\sqrt{5}}{4} = .559$$

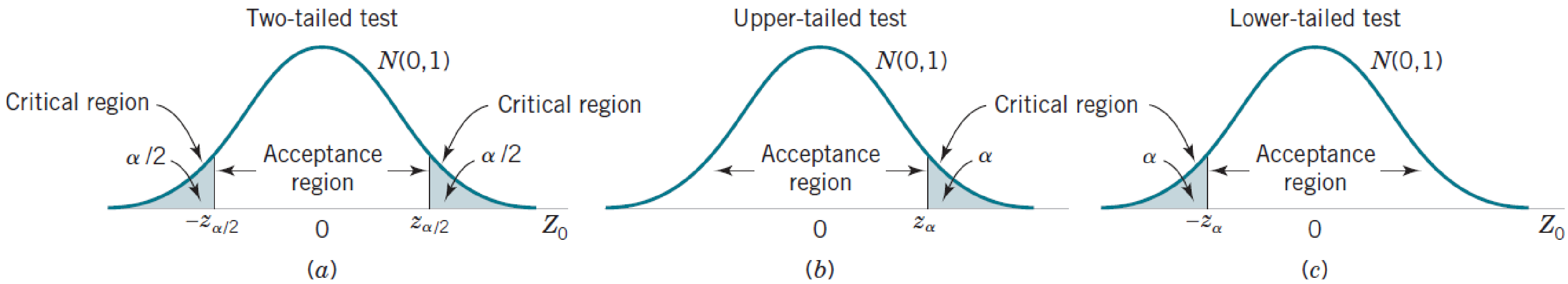
$$\begin{aligned} P\{|Z| > .559\} &= 2P\{Z > .559\} \\ &= 2 \times .288 = .576 \end{aligned}$$

p-value is .576

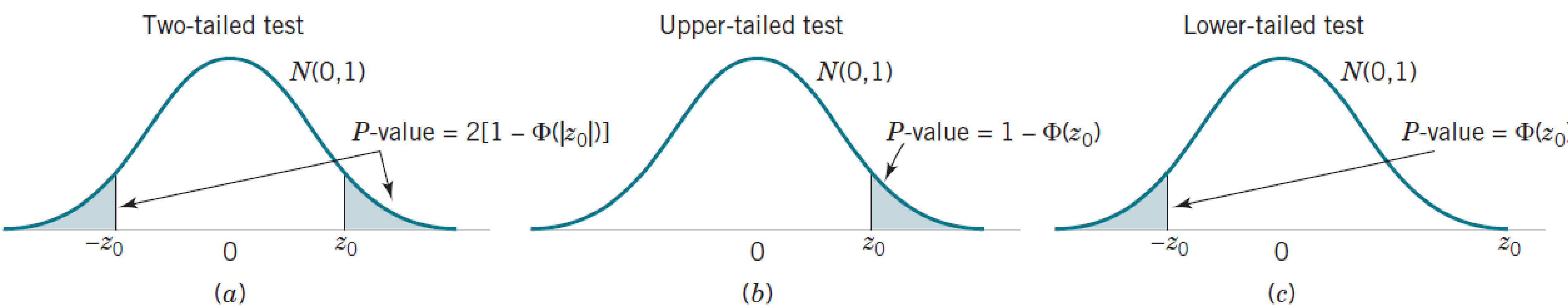
Table 8.1 $X_1, \ldots, X_n$ Is a Sample from a (μ, σ^2) Population, σ^2 Is Known, $\overline{X} = \sum_{i=1}^{n} X_i/n$.

H_0	H_1	Test Statistic TS	Significance Level α Test	p-Value if $TS = t$
$\mu = \mu_0$	$\mu \neq \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $\|TS\| > z_{\alpha/2}$	$2P\{Z \geq \|t\|\}$
$\mu \leq \mu_0$	$\mu > \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $TS > z_\alpha$	$P\{Z \geq t\}$
$\mu \geq \mu_0$	$\mu < \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $TS < -z_\alpha$	$P\{Z \leq t\}$

Z is a standard normal random variable.

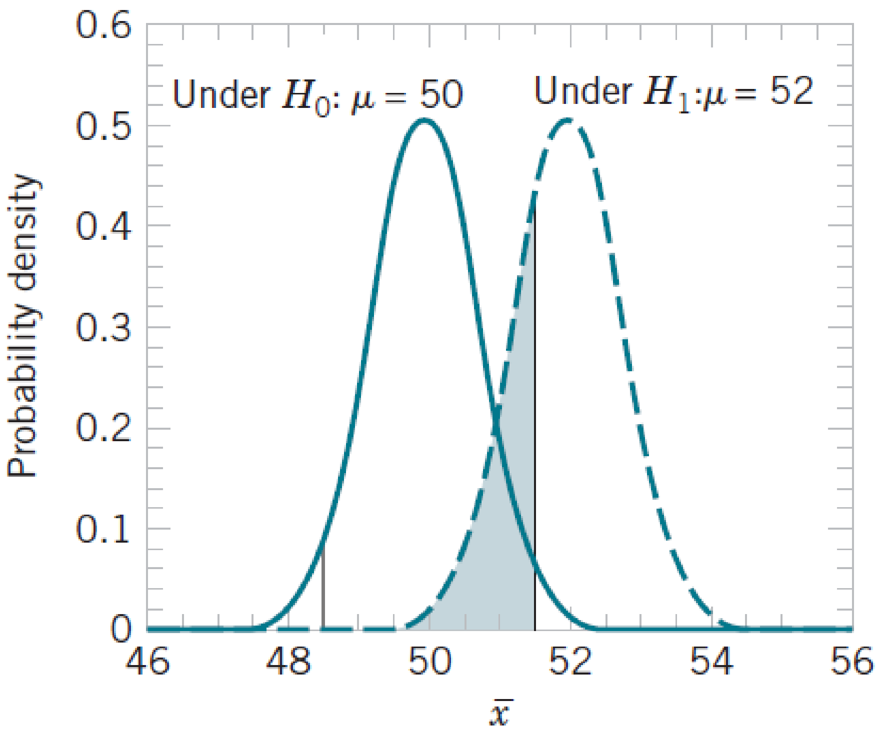


Two-tailed test
N(0,1)
Critical region
Critical region
α /2
α /2
Acceptance region
−z α/2
0
z α/2
Z0
(a)
Upper-tailed test
N(0,1)
Critical region
Acceptance region
α
0
z α
(b)
Lower-tailed test
N(0,1)
α
Acceptance region
−z α
0
Z0
(c)



Two-tailed test
N(0,1)
P-value = 2[1 − Φ(|z0|)]
−z0
0
z0
(a)
Upper-tailed test
N(0,1)
P-value = 1 − Φ(z0)
0
z0
(b)
Lower-tailed test
N(0,1)
P-value = Φ(z0)
−z0
0
z0
(c)

Probability of Type II error



$$= \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)$$

EXAMPLE 8.3c For the problem presented in Example 8.3a, let us determine the probability of accepting the null hypothesis that $\mu = 8$ when the actual value sent is 10. To do so,

EXAMPLE 8.3e Suppose in Example 8.3a that we know in advance that the signal value is at least as large as 8. What can be concluded in this case?

SOLUTION To see if the data are consistent with the hypothesis that the mean is 8, we test

$$H_0 : \mu = 8$$

against the one-sided alternative

$$H_1 : \mu > 8$$

The value of the test statistic is $\sqrt{n}(\overline{X} - \mu_0)/\sigma = \sqrt{5}(9.5 - 8)/2 = 1.68$, and the p-value is the probability that a standard normal would exceed 1.68, namely,

$$p\text{-value} = 1 - \Phi(1.68) = .0465$$

Since the test would call for rejection at all significance levels greater than or equal to .0465, it would, for instance, reject the null hypothesis at the $\alpha = .05$ level of significance. ■

$$\frac{\sqrt{n}}{\sigma}(\mu_0 - \mu) = -\frac{\sqrt{5}}{2} \times 2 = -\sqrt{5}$$

As $z_{.025} = 1.96$, the desired probability is, from Equation 8.3.4,

$$\begin{aligned}
&\Phi(-\sqrt{5} + 1.96) - \Phi(-\sqrt{5} - 1.96) \\
&\quad = 1 - \Phi(\sqrt{5} - 1.96) - [1 - \Phi(\sqrt{5} + 1.96)] \\
&\quad = \Phi(4.196) - \Phi(.276) \\
&\quad = .392 \quad \blacksquare
\end{aligned}$$

The function $1 - \beta(\mu)$ is called the *power-function* of the test. Thus, for a given value μ, the power of the test is equal to the probability of rejection when μ is the true value. $\blacksquare$

$$n \approx \frac{(z_{\alpha/2} + z_{\beta})^2 \sigma^2}{(\mu_1 - \mu_0)^2}$$

Example 8.3.d. For the problem of Example 8.3.a, how many signals need be sent so that the .05 level test of $H_0 : \mu = 8$ has at least a 75 percent probability of rejection when $\mu = 9.2$?

Solution. Since $z_{.025} = 1.96$, $z_{.25} = .67$, the approximation from Equation (8.3.7) yields

$$n \approx \frac{(1.96 + .67)^2}{(1.2)^2} 4 = 19.21$$

Hence a sample of size 20 is needed. From Equation (8.3.4), we see that with $n = 20$

$$\begin{aligned}
\beta(9.2) &= \Phi\left(-\frac{1.2\sqrt{20}}{2} + 1.96\right) - \Phi\left(-\frac{1.2\sqrt{20}}{2} - 1.96\right) \\
&= \Phi(-.723) - \Phi(-4.643) \quad \approx 1 - \Phi(.723) \\
&\qquad\qquad\qquad\qquad\qquad\qquad \approx .235
\end{aligned}$$

Therefore, if the message is sent 20 times, then there is a 76.5 percent chance that the null hypothesis $\mu = 8$ will be rejected when the true mean is 9.2. ■

Case of unknown Variance: the t-test

Table 8.2 $X_1, \ldots, X_n$ Is a Sample from a (μ, σ^2) Population, σ^2 Is Unknown, $\overline{X} = \sum_{i=1}^{n} X_i / n S^2 = \sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)$.

H_0	H_1	Test Statistic TS	Significance Level α Test	p-Value if $TS = t$
$\mu = \mu_0$	$\mu \neq \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/S$	Reject if $\|TS\| > t_{\alpha/2,\, n-1}$	$2P\{T_{n-1} \geq \|t\|\}$
$\mu \leq \mu_0$	$\mu > \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/S$	Reject if $TS > t_{\alpha,\, n-1}$	$P\{T_{n-1} \geq t\}$
$\mu \geq \mu_0$	$\mu < \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/S$	Reject if $TS < -t_{\alpha,\, n-1}$	$P\{T_{n-1} \leq t\}$

T_{n-1} is a t-random variable with $n-1$ degrees of freedom: $P\{T_{n-1} > t_{\alpha,\, n-1}\} = \alpha$.

Example 8.3.i. The manufacturer of a new fiberglass tire claims that its average life will be at least 40,000 miles. To verify this claim a sample of 12 tires is tested, with their lifetimes (in 1000s of miles) being as follows:

Tire	1	2	3	4	5	6	7	8	9	10	11	12
Life	36.1	40.2	33.8	38.5	42	35.8	37	41	36.8	37.2	33	36

Test the manufacturer's claim at the 5 percent level of significance.

Solution. To determine whether the foregoing data are consistent with the hypothesis that the mean life is at least 40,000 miles, we will test

$$H_0 : \mu \geq 40{,}000 \qquad \text{versus} \qquad H_1 : \mu < 40{,}000$$

A computation gives that

$$\overline{X} = 37.2833, \qquad S = 2.7319$$

and so the value of the test statistic is

$$T = \frac{\sqrt{12}(37.2833 - 40)}{2.7319} = -3.4448$$

Since this is less than $-t_{.05,11} = -1.796$, the null hypothesis is rejected at the 5 percent level of significance. Indeed, the p-value of the test data is

$$p\text{-value} = P\{T_{11} < -3.4448\} = P\{T_{11} > 3.4448\} = .0027$$

Testing the equality of means of two normal populations

Table 8.4 $X_1, \ldots, X_n$ Is a Sample from a (μ_1, σ_1^2) Population; $Y_1, \ldots, Y_m$ Is a Sample from a (μ_2, σ_2^2) Population,
The Two Population Samples Are Independent to Test
$H_0 : \mu_1 = \mu_2$ versus $H_0 : \mu_1 \neq \mu_2$.

Assumption	Test Statistic TS	Significance Level α Test	p-Value if $TS = t$
σ_1, σ_2 known	$\dfrac{\overline{X}-\overline{Y}}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}}$	Reject if $\lvert TS \rvert > z_{\alpha/2}$	$2P\{Z \geq \lvert t \rvert\}$
$\sigma_1 = \sigma_2$	$\dfrac{\overline{X}-\overline{Y}}{\sqrt{\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}\sqrt{1/n + 1/m}}$	Reject if $\lvert TS \rvert > t_{\alpha/2,\, n+m-2}$	$2P\{T_{n+m-2} \geq \lvert t \rvert\}$
n, m large	$\dfrac{\overline{X}-\overline{Y}}{\sqrt{S_1^2/n + S_2^2/m}}$	Reject if $\lvert TS \rvert > z_{\alpha/2}$	$2P\{Z \geq \lvert t \rvert\}$

Treated with Vitamin C	Treated with Placebo
5.5	6.5
6.0	6.0
7.0	8.5
6.0	7.0
7.5	6.5
6.0	8.0
7.5	7.5
5.5	6.5
7.0	7.5
6.5	6.0
	8.5
	7.0

Do the data listed prove that taking 4 grams daily of vitamin C reduces the mean length of time a cold lasts? At what level of significance?

SOLUTION To prove the above hypothesis, we would need to reject the null hypothesis in a test of

$$H_0 : \mu_p \leq \mu_c \qquad \text{versus} \qquad H_1 : \mu_p > \mu_c$$

where μ_c is the mean time a cold lasts when the vitamin C tablets are taken and μ_p is the mean time when the placebo is taken. Assuming that the variance of the length of the cold is the same for the vitamin C patients and the placebo patients, we test the above by running Program 8.4.2. This yields the information shown in Figure 8.6. Thus H_0 would be rejected at the 5 percent level of significance.

$$\overline{X} = 6.450, \qquad \overline{Y} = 7.125$$

$$S_x^2 = .581, \qquad S_y^2 = .778$$

$$S_p^2 = \frac{9}{20} S_x^2 + \frac{11}{20} S_y^2 = .689$$

$$TS = \frac{-.675}{\sqrt{.689(1/10 + 1/12)}} = -1.90$$

Since $t_{0.5,20} = 1.725$, the null hypothesis is rejected at the 5 percent level of significance.

The paired t-test

n pairs $(X_i, Y_i), i = 1, \ldots, n,$

gas consumption of the ith car

X_i Y_i

cannot treat $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$ are independent samples

$$\text{let } W_i = X_i - Y_i,\ i = 1, \ldots, n.$$

$$H_0 : \mu_w = 0 \quad \text{versus} \quad H_1 : \mu_w \neq 0$$

where $W_1, \ldots, W_n$ are assumed to be a sample from a normal population having unknown mean μ_w and unknown variance σ_w^2. But the t-test described in

$$\text{accepting } H_0 \text{ if } -t_{\alpha/2,\,n-1} < \sqrt{n}\frac{\overline{W}}{S_w} < t_{\alpha/2,n-1}$$

$$\text{rejecting } H_0 \text{ otherwise}$$

EXAMPLE 8.4d An industrial safety program was recently instituted in the computer chip industry. The average weekly loss (averaged over 1 month) in labor-hours due to accidents in 10 similar plants both before and after the program are as follows:

Plant	Before	After	$A - B$
1	30.5	23	−7.5
2	18.5	21	2.5
3	24.5	22	−2.5
4	32	28.5	−3.5
5	16	14.5	−1.5
6	15	15.5	.5
7	23.5	24.5	1
8	25.5	21	−4.5
9	28	23.5	−4.5
10	18	16.5	−1.5

Determine, at the 5 percent level of significance, whether the safety program has been proven to be effective.

$$H_0 : \mu_A - \mu_B \geq 0 \qquad \text{versus} \qquad H_1 : \mu_A - \mu_B < 0$$

because this will enable us to see whether the null hypothesis that the safety program has not had a beneficial effect is a reasonable possibility. To test this, we run Program 8.3.2, which gives the value of the test statistic as -2.266, with

$$p\text{-value} = P\{T_q \leq -2.266\} = .025$$

Since the p-value is less than .05, the hypothesis that the safety program has not been effective is rejected and so we can conclude that its effectiveness has been established (at least for any significance level greater than .025). ■

Hypothesis tests concerning the variance

Let $X_1, \ldots, X_n$ denote a sample from a normal population unknown mean μ and unknown variance σ^2

test the hypothesis

$$H_0 : \sigma^2 = \sigma_0^2$$

versus the alternative

$$H_1 : \sigma^2 \neq \sigma_0^2$$

$$\frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2_{n-1}$$

Because $P\{\chi^2_{n-1} < \chi^2_{\alpha/2,n-1}\} = 1 - \alpha/2$ and $P\{\chi^2_{n-1} < \chi^2_{1-\alpha/2,n-1}\} = \alpha/2$, it follows that

$$P_{H_0}\left\{\chi^2_{1-\alpha/2,n-1} \leq \frac{(n-1)S^2}{\sigma_0^2} \leq \chi^2_{\alpha/2,n-1}\right\} = 1 - \alpha$$

Therefore, a significance level α test is to

$$\text{accept } H_0 \quad \text{if} \quad \chi^2_{1-\alpha/2,n-1} \leq \frac{(n-1)S^2}{\sigma_0^2} \leq \chi^2_{\alpha/2,n-1}$$

$$\text{reject } H_0 \quad \text{otherwise}$$

$$p\text{-value} = 2\min(P\{\chi^2_{n-1} < c\}, 1 - P\{\chi^2_{n-1} < c\})$$